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Studierichting: Informatica
Oefeningenreeks 5A nr. 11.4

$$\int_0^1 \left(x^2+3x+2\right)e^x dx$$

$$\text{O.I.} = \int \left(x^2+3x+2\right) d\left(e^x\right)$$

$$= \left(x^2+3x+2\right)e^x - \int (2x+3)\, d\left(e^x\right)$$

$$= \left(x^2+3x+2\right)e^x - (2x+3)\, e^x + \int 2e^x dx$$

$$= \left(x^2+3x+2-2x-3+2\right)e^x + c = \left(x^2+x+1\right)e^x + c$$

$$\Rightarrow \text{B.I.} = -\left[\left(x^2+x+1\right)e^x\right]_{-1}^{0} = -\left(1-\frac{1}{e}\right) = \frac{1-e}{e}$$

References